

The Effect of Time Separation on Coordination Processes and Outcomes: A Case Study

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Abstract

Prior research has discussed and investigated how global teams bridge multiple boundaries (e.g., distance, time, culture) that separate its members and impact the coordination of their work. In this paper we report on a study we conducted at a semiconductor manufacturing company to better understand how global teams can work more effectively across one of these boundaries – time separation. More specifically, we investigate how time zones affect coordination costs and other coordination outcomes, and which coordination processes and mechanism are more effective in helping overcome the difficulties associated with time separation. The study was conducted through semi-structured interviews of 23 global team members working on technical projects in several locations around the globe.

1. Introduction

Work is becoming increasingly more global with team members spread across multiple locations and time zones. These global teams need to bridge multiple internal boundaries (e.g., geographic, time zones, language, culture, etc.) to get the job done [9, 19, 29, 36]. This can represent substantial problems for technical project teams in which members' task activities are highly interdependent, thus requiring substantial coordination [34], which has been documented in global technical tasks like software development [14]. However, the very presence of these team boundaries not only reduces the range of coordination mechanisms available to team members, but also makes it more difficult to employ the available tools and mechanisms, thus making them less effective. When team members are separated by time and distance, their interaction is largely mediated by leaner communication and information technologies [6, 7], which reduces the benefits of co-presence (e.g., frequent and spontaneous interaction, availability of contextual reference, presence awareness, workspace awareness, etc.). In essence, we argue that time differences affect coordination outcomes and tend to increase coordination costs.

While much has been written about coordination in global teams [3, 13, 15] and geographically dispersed teams in general [1, 4, 16, 17, 20, 24, 28, 30, 35], none of this work, as far as we know, has isolated the effect of the individual boundaries. Geographic dispersion often co-varies with other boundaries, thus making it difficult to isolate the effects of a particular boundary in prior research. We are particularly interested in better understanding the effects of time separation on coordination outcomes. A recent review of the virtual team literature identified the role of time in virtual teams as an important area of research, which has not been adequately investigated. Espinosa and Carmel [8] articulated and proposed a conceptual model that describes how time separation affects coordination costs, but to of our knowledge, no empirical work has been done to validate this model or to the effects of time coordination on other outcomes. We try to fill this gap in this study based on semi-structured interviews conducted at one semiconductor organization.

Distance and time separation are most important when studying dispersed teams mediated by technology [2], but while distance separation has received a lot of attention, time separation has gone largely neglected. We argue in this paper that, with today's availability of sophisticated collaboration technology and teams' increasing experience working globally, geographic distance is no longer a substantial problem for teams as it used to be, whereas time separation remains a great challenge, particularly as more time zones are represented in a team.

In this study we address the following research questions: (1) *what is the effect of time separation on coordination outcomes and costs in global teams?* And, (2) *which coordination processes, mechanisms and tools help overcome the difficulties of time separation?* In addition, we do some preliminary empirical validation of the time separation and coordination costs model proposed by Espinosa and Carmel [8]. The next section of this paper discusses our study model followed by the methodology we employed for the study. We then present our findings and corresponding propositions. Next, we discuss the

implications of our finding for solution design, followed by our concluding remarks.

2. Time Separation and Coordination: Study Framework

Our research framework follows the typical “input-process-outcome” (I-P-O) model widely used to study effectiveness in groups that interact with technology [25] and for virtual teams [5, 30]. We describe our framework below, starting with coordination outcomes (O), followed by coordination processes and mechanisms (P), and input variables, which we refer to as context or situational variables (S).

2.1 Coordination Outcomes (O)

Coordination theory defines coordination as the effective management of dependencies among activities in the task [22]. Basically, if two people are working together in the task and their respective activities can be carried out independently, there is nothing to coordinate. Conversely, if their activities are highly interdependent, then the respective dependencies have to be effectively managed, often creating substantial coordination overhead. Two coordination outcomes of interest in this research are coordination effectiveness and coordination costs. Coordination effectiveness is very important for performance in tasks with substantial dependencies [33, 34]. Consistent with this definition, we define coordination effectiveness as the resulting outcome in which all key dependencies among activities in a task have been well managed (e.g., software parts integrated well, all parts were completed on time, there were no priority conflicts, etc.).

While coordination effectiveness is beneficial for team performance, coordination activities divert attention from productive activities. So, it is possible to have a well coordinated team with poor performance because of the substantial coordination costs incurred in managing dependencies. Coordination costs are the tangible or intangible costs incurred in managing task dependencies, including extra effort [5], communication and vulnerability costs due to misunderstandings [21]. In their conceptual model, Espinosa and Carmel [8] argued that total implementation costs for geographically dispersed team projects can be broken down into individual production costs and coordination costs, and that coordination costs can be broken down into four components: (1) pure communication costs – i.e., the cost of composing and sending messages, plus the cost of installation and use of communication mechanisms; (2) pure cost of delay – measured from the point of view of “my site” which is waiting for a response (i.e., delay while my site works and the other site sleeps is increased; delay while my site sleeps and the other site works is zero); (3) clarification costs –

i.e., additional communication and delay necessary to repair misunderstandings; and (4) re-work costs – i.e., additional production costs when misunderstanding lead to re-work. The model predicts that time separation: (1) reduces pure communication costs because asynchronous communication technologies are less expensive than synchronous communication technologies; (2) reduces or increases pure delay costs, depending on the timing and organization of activities – i.e., well timed and synchronized workflows can use time zone differences advantageously by handing over work to other sites at the end of a work day; (3) increases clarification costs; and (4) increases re-work costs. Generally speaking, preliminary analysis of the results of the present study strongly supports the above model.

2.2 Coordination Processes and Mechanisms (P)

A coordination process is a set of actions implemented to manage task dependencies. Similarly, a coordination mechanism is a procedure or tool used to manage task dependencies. We use these two terms interchangeably because we view the use of coordination mechanisms as a coordination process. The classic organizational research literature suggest that teams use two main kinds of mechanisms to coordinate their work: task programming and communication [23, 33, 34]. Task programming mechanisms are things like plans, specifications, schedules, and tools, which are effective to manage the most routine task dependencies. Communication is most effective when managing less routine task dependencies. Communication is also necessary when a routines change (e.g., system breakdowns, schedule delays).

At the same time, the team cognition research literature suggest that members also use cognitive mechanisms like shared knowledge and team awareness to manage their dependencies more effectively. When teams share knowledge about the team or the task, this knowledge helps them anticipate what others are likely to do and what the task requires from them (Cannon-Bowers, Salas et al. 1993; Klimoski and Mohammed 1994), thus helping them coordinate implicitly. In addition, when team members know that they share knowledge, their communication is more effective because they have more common ground (Krauss and Fussell 1990; Cramton 2001). Similarly, team awareness helps members coordinate because it provides contextual information for their individual activities (Dourish and Bly 1992). It provides up to the minute perception and comprehension of what is happening in the shared task environment, which enables the team to act in a synchronized fashion (Wellens 1993; Endsley 1995).

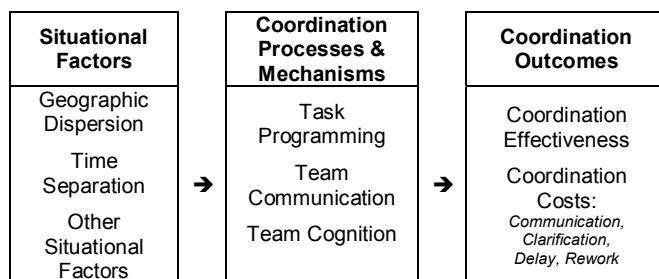
2.3 Situational Factors (S)

We use the term “situational factors” to group all relevant input variables related to the work context, task

characteristics and other related factors describing the situation in which the team operates. Situational factors can be numerous, including team boundaries (e.g., geographic, time zones, language, culture, etc.), nature of the task (e.g., types of dependencies, type of task), characteristics of the team (e.g., team members experience, prior experience working with each other, prior experience working in global projects, etc.). We pay particular attention to two situational variables of interest for this study: distance separation and time separation. While the focus of our study is on time separation, distance separation co-varies substantially with time separation. Therefore, we try to distinguish between co-located collaboration, north-south collaboration (separated, same time zone) and east-west collaboration (separate, different time zone) [27].

Prior research [8] suggest that: (1) time separation is more than just time zones – it includes differences in holidays, weekends, work schedules, etc.; (2) time separation leads to substantial changes in work norms – e.g., work hours – and various tactics – e.g., redundant roles, expanded overlap windows, team training, 24-hour availability, etc. – to cope with this challenge; (3) during non-overlapping hours, time separated team members have to make choices between communicating asynchronously (imperfectly) or waiting for an overlapping time to communicate synchronously; (4) distance separation is symmetric whereas time separation is asymmetric – i.e., it matters if the other site is ahead or behind your local time; (5) differences in how team members are configured across time zones makes a big difference; (6) time perspectives are the same when members are separated by distance only, but not when they are separated by time – i.e., one site works while the other sleeps; (7) vulnerability costs – i.e., additional cost/delays due to misunderstandings and/or re-work – increases with time separation.

Figure 1: Research Model for Time Separation and Coordination Costs in Global IS Projects



In sum, team members have a wide range of coordination mechanisms, traditional and cognitive, that they can use to manage their dependencies. We anticipate that certain characteristics of the work context will require a different mix of coordination mechanisms than others. When the situational factors in a work context are such that certain

coordination mechanism are not as effective or are not readily available, we expect that team members will use other mechanisms that are more effective. In this study, we argue that time separation has a strong effect on how and when team members communicate. Therefore, task programming mechanisms can become an important substitute. At the same time, task programming mechanisms are not as effective under uncertain and less routine conditions, thus the importance of team cognition. However, this perspective – i.e., understanding how time separation affects coordination effectiveness and coordination costs, and which coordination mechanisms are most effective under different conditions of time separation – has not been empirically investigated. Our study aims at bridging this gap. The research model for this study is illustrated in figure 1.

3. Study Method

This study was conducted via semi-structured telephone interviews of 23 team members working on technical projects for a semiconductor manufacturing company in several global locations, including Colorado, California, New Mexico, Oregon, Utah, India, Vancouver, England, Ireland, Philippines, Israel, Russia, and China. Interviews were approximately one hour long and they included questions aimed at contrasting responses across co-located, north-south and east-west collaborations.

In most of the questions, participants were asked to contrast their responses across co-located, distance-only and time-separated contexts. Participants were first asked general questions, without focusing on any particular project yet, intended to elicit general statements about how situational variables – particularly time separation – affected coordination outcomes ($S \rightarrow O$). The next set of questions focused more specifically on a current or recent project that the respondent had worked or is working on. This set of questions was intended to elicit statements about how situational variables – particularly time separation – influenced their team processes ($S \rightarrow P$) and how these team processes affected coordination outcomes ($P \rightarrow O$). Participants were asked specific questions about the communication, task programming and cognitive processes and mechanisms they used to coordinate their work, contrasting the use and effectiveness of these processes and mechanisms across co-located, distance-only and time-separated contexts. All interviews were audio taped with the participants' consent after we gave them assurances of anonymity and confidentiality. The interviews were transcribed verbatim by an external service provider, yielding 354 single-spaced pages of interview transcriptions, containing 147,190 words.

The method used to code and analyze the data is based on grounded theory [12, 32]. We first developed a coding template with very general themes of interest (e.g., time separation, coordination success, delay). As prescribed by grounded theory, we did “open coding” of the text data to verify that these themes were relevant, identify new themes, and eliminate themes not discussed by the participants. We then did axial coding of the data to create a hierarchical coding scheme using these themes of interest as the starting high level codes, and then linking to them the sub-codes that emerged from the data. Hierarchical coding schemes are useful for qualitative analysis because they facilitate aggregation of results to high level themes, while retaining fine grained level of detail in the sub-codes. We used higher level codes (e.g., S, O and P) representing main theme categories and lower level themes (e.g., P.Cm.S.F) representing the specific details identified in the theme (e.g., a [P] process of [Cm] communication involving [S] synchronous interaction [F] face-to-face).

We then did further “axial coding” of the interview text data to uncover lateral links across themes. The objective on this part of the coding was to find cause-effect relationships among code categories (e.g., time separation causes delay), which is consistent with other similar methods of qualitative data analysis like codebook analysis or thematic coding [26] and attributional coding, which seeks to identify cause-effect attributions made by participants [31]. While we coded all attributions made by participants and the relations they described among themes, we were most interested in three types of attributions: ($S \rightarrow O$) i.e., which situational factors affected which outcomes – more specifically, we were interested in how time separation affected coordination success and coordination costs (e.g., time separation causes delay); ($S \rightarrow P$) which situational factors affected the use of specific coordination processes or mechanisms – more specifically, we were interested in how time separation affected the use of communication, task programming and cognitive coordination mechanisms (e.g., substantial time zone differences makes us rely more heavily on formal documents); and ($P \rightarrow O$) what effect did the use of these processes or mechanisms have on which outcomes – again, we were interested in how the use of various communication, task programming and cognitive coordination mechanisms affected coordination success and coordination costs (e.g., knowing who knows what in the team makes finding information more effective).

All segments of the coded text that had meaning for this research were extracted into a database for easy retrieval and sorting of attributions by code. A total of 1125 unique coded segments of text were extracted to the database for analysis. After the coding was completed, the text was given to an external coder – a graduate student with substantial working experience. The external coder was

trained on the coding scheme, who then coded five interview transcripts. The disagreements with the external coder were then discussed as part of the training. Because the level of agreements for these five interviews was quite high at this point, the external coder was instructed to code the remaining interviews. The resulting reliability was 72.4%, measured as the ratio of agreements to agreements plus disagreements [26], indicating substantial agreement [18]. The results reported in this study are based on qualitative interpretations of the text segments we extracted and coded. As prescribed by grounded theory, the text was analyzed by comparing similarities and differences in the data, first by reading the cases in sequential order, and then reading the coded text sorted by themes. The main comparisons were made between co-located, distance-only, and time-separated contexts.

4. Findings

We first discuss the situational (S), process (P) and outcome (O) themes we observed. We then discuss our findings about relational attributions ($S \rightarrow O$, $S \rightarrow P$ and $P \rightarrow O$). Throughout this section, we formulate the corresponding propositions whenever the findings pertain to time separation.

4.1 Situational Factors (S)

Consistent with prior studies [27], technical teams in this organization work in various geographic configurations, including: (1) Co-located – everyone works in the same location; (2) North-South – 2 or more locations in the same time zone; (3) East-West or Follow the Sun – 2 or more locations with substantial time-zone difference and little or no time overlap; (4) Dispersed balanced – team members widely scattered across multiple time zones; (5) Dispersed imbalanced – majority of team members located in core site with a few other members in various sites across multiple time zones. We found that these configurations have a substantial effect on the coordination challenges faced by the teams. Consistent with arguments made by researchers of team boundaries [9, 29, 36], the more geographic and time boundaries exist within a team, the more severe the coordination challenges.

Proposition 1: As more time zones are represented in a team, the coordination challenges increase.

Co-located team members had an average of 4.1 years of prior experience with each other, whereas distant and time-separated members had 1.6 and 1.4 years respectively. Co-located members often cited their prior knowledge of team members as a positive factor. Similarly, distant and time-separated members who met face-to-face before the project to get to know each other, learn about each others’ interaction styles and organize their work processes, viewed

this as very favorable, which is consistent with prior research findings [16].

Proposition 2: Prior experience with team members helps cope with coordination challenges across time zones.

When team members collaborate with other members, they have dependencies among their respective activities, which they need to manage to keep things coordinated [22]. Consistent with prior studies of coordination in technical teams [10], the nature of dependencies observed varied as: (1) technical dependencies – i.e., integrating different parts, exchanging technical details; (2) temporal dependencies – i.e., completing individual tasks on schedule to stay synchronized with other team members; and (3) process dependencies – i.e., understanding and following established procedures, priorities, etc. However, most dependencies discussed were technical and these dependencies were more salient across time-separated members, followed by distance-only members, and then by co-located members. Process dependencies were next in relevance and there was no difference across collaboration modes. Interestingly, temporal dependencies were the least salient of the three.

Proposition 3: The type of coordination challenges faced by a team will vary depending on the nature of dependencies (i.e., technical, temporal or process) among their task activities and these dependencies are more salient to time-separated teams.

4.2 Coordination Processes and Mechanisms (P)

Team members use a variety of synchronous communication mechanisms to coordinate their work. There is no clear preference among these mechanisms (i.e., face-to-face, telephone, instant messenger and web data conferencing—NetMeeting™), except that: (1) face-to-face communication is the preferred method in co-located contexts, while it is not available in other contexts and (2) video conference is not widely used with non-co-located members. Other than this, the method of communication used varied depending on personal preferences, task priority and the probability of finding the other team member. In contrast, team members use very few asynchronous communication mechanisms. By far, the most predominant asynchronous communication is electronic mail. Some team members talked about using technical web sites, while no one mentioned web logs, discussion groups, or shared databases (except for special collaboration tools discussed below).

Team members use a variety of task programming mechanisms. Most participants discussed the use of: (1) centralized documents to access process, project or product information ; (2) task organization arrangements – e.g.,

division of labor, redundant roles across sites, re-location of personnel, meeting schedules, and special procedures to deal with time zones; (3) use of special collaboration tools – e.g. Share Point™, Outlook™ shared calendar, shared drives, software tools, etc. Team members also use various types of cognitive mechanisms. Task awareness (i.e., knowledge of who has done what with respect to the task and when) was the cognitive mechanism most often mentioned. Transactive memory (i.e., knowing who knows what in the team) was the next most salient cognitive mechanism for most team members. Some form of shared knowledge task or about each other and presence awareness were also salient for most team members. While not cited by the majority, a few members indicated that having shared beliefs (e.g., shared goals, common vision about the project, etc.) was also helpful.

4.3 Outcomes (O)

Consistent with the Espinosa-Carmel model [8], we observed that coordination costs were very salient to most team members, particularly those associated with delay, which was often associated with distance and time separation. The next most salient coordination cost issue, also mostly associated with distance and time separation, was effort, which manifests itself in many ways, including: extended or adjusted work schedules, round-the-clock availability, additional documentation and communication, and additional travel and training. The next most salient issue discussed by little over half of the respondents was the cost of repairing miscommunication and only very few respondents discussed the need for rework because of this miscommunication.

Proposition 4: Coordination costs – i.e., delay, effort, and miscommunication – are very salient to time-separated team members.

With respect to coordination effectiveness, interestingly, while temporal dependencies were the least salient when participants discussed their dependencies with other team members (see S findings above), it was the most salient to most team members when describing coordination effectiveness, followed by process and technical dependencies.

4.4 The Effect of Situational Factors on Outcomes (S→O)

Most team members felt that their work was well coordinated or somewhat coordinated. While co-located team members were more coordinated than distant members, and distant members were more coordinated than time-separated members, the difference was very small. This is perhaps due to the fact that corporate pressures and competitiveness require most projects to be successful, but

as we discuss in the next finding, coordination effectiveness in time-separated contexts is only achieved with substantial additional coordination costs. This finding suggests that while time proximity reduces the effort necessary to coordinate, this doesn't necessarily mean that teams will be better coordinated in the end. It appears that when the coordination barriers imposed by time separation are removed, teams experience other problems (e.g., meeting overload, working on multiple projects, production losses—too many one-on-one face-to-face meetings reduce available production times).

Proposition 5: For teams driven by high performance standards, there is negligible difference in coordination effectiveness with time separation.

While most projects were successfully coordinated, time separation required a substantial amount of additional effort by team members. This effort was materialized in several forms, including: extraordinary effort required to schedule meetings; adjusting and extending work schedules to increase overlap time between sites; the need to work from home and be available almost around the clock; creating redundancies and overlapping responsibilities between sites; additional meetings and training sessions at the beginning of the project; the need for special and more rigorous work processes, task organization and team interaction routines; and the need to get additional information and more details upfront before meetings. This finding is consistent with prior research [5], but we also found that this additional effort required is greatly increased as more time zones are involved in a project. Whereas a team with only two time zones can make adjustment to work practices and schedules to cope with reduced overlapping times, this becomes a monumental challenge when team members are spread out across several time zones. In one account, participants shifted their schedules so much that co-located team members rarely saw each other (i.e., co-located time separation), suggesting that time separation needs to be measured based on actual work schedules, and not so much based on geographic time zone differences.

Proposition 6: Time separation increases coordination costs –i.e., it leads to substantial additional effort and this effort increases as time separation increases.

Similarly, the majority of participants indicated that time separation lead to higher coordination costs in the form of increased delay. Consistent with the Espinosa-Carmel model, when delay is measured from the perspective of the person waiting for a response, pure delay (i.e., the time it takes to receive a response) is sometimes increased and sometimes decreased with time zones. Some participants discussed situations in which time zones caused more response delay (e.g., need a response right now but the

other site is off working hours). However some participants also indicated that time zones reduced delay (e.g., work was sent off to a remote site at the end of the day and the work was ready first thing next day). However, also consistent with the model, almost half of the participants indicated that having to repair misunderstandings across time zones had a severe impact on delay and that it takes an extra day in the response cycle for a misunderstanding to be notified and corrected, which doesn't happen in same time zone contexts. A few participants also indicated that such misunderstandings often led to the need to do re-work. At the same time, most participants, when asked to contrast outcomes with same-time zone distributed work and co-located work, they clearly suggested that when work overlap is maximized it is easier to coordinate the work and that coordination costs were lowered.

Proposition 7: Time separation increases coordination costs – i.e., pure delay may increase or decrease depending on the timing of workflow across sites, but total delay increases substantially when teams need to interact frequently and when there is a need to repair misunderstandings.

4.5 The Effect of Situational Factors on the Use of Coordination Processes and Mechanisms (S→P)

When team members are separated by distance, their communication options narrow, but most team members have experience working globally and have been able to adjust to distance separation. Furthermore, the availability of synchronous collaboration tools like web data conferencing (e.g., NetMeeting™, WebX™, instant messenger, pagers, mobile phones) have reduced the relevance of geographic distance as a factor. Furthermore, when other asynchronous tools (e.g., Outlook™ calendars) are used by these teams, their members enjoy most of the benefits of co-location (e.g., presence awareness, spontaneous interaction, task awareness, and rich communication media). This is confirmed by our findings, which suggest that distance separation does not appear to be a critical factor affecting team coordination at this company. Time separation, on the other hand, does present monumental challenges to these teams.

Interestingly, team members develop preferences for the use of coordination mechanisms from their own prior experience and according to their interaction style. Once they adopt certain interaction routines and coordination mechanisms, they tend to use them for all contexts, except when the context imposes restrictions in their use (e.g., distant members can't meet face to face, time-separated members can't communicate synchronously during non-overlapping hours). For example, when asked to contrast the differences in the use of coordination mechanisms across collaboration modes, the response was often "it is the

same thing, but it is easier in the same time zone or co-located". Team members tend to gravitate to the coordination mechanisms they prefer the most, which is often face-to-face communication. However, as discussed earlier, if face-to-face communication is not the optimal coordination method for a particular type of task (i.e., some highly technical tasks require substantial exchange of written material), this can lead to substantial production losses. Coordination theory suggests that direct communication is more effective for non-routine and more equivocal aspects of the task, but co-located members tend to use more face-to-face interaction, which promotes social interaction and stronger relationships.

Proposition 8: Preferences for which coordination mechanisms to use are generally unaffected by time separation – i.e., teams may adopt certain coordination mechanisms to cope with time separation, but once adopted they tend to be used in all collaboration modes.

While preferences for coordination mechanisms may not change with distance or time separation, the availability of these mechanisms is indeed affected. Co-located team members have the full array of coordination mechanisms to their disposal, the range of available coordination mechanisms narrows when the team is separated by distance or time. Geographically separated teams have fewer coordination mechanisms available than co-located teams. Similarly, time separated teams have fewer coordination mechanisms available than teams in the same time zone, but time separation has the added complication that the coordination mechanisms available to the team change with the time of the day. Therefore, team members need to make decisions on whether to communicate asynchronously or wait until work hours overlap and then communicate synchronously, which is consistent with prior research [8]. Furthermore, as more time zones are represented in a team, the coordination challenges increase and it becomes quite difficult to find a window of time when work hours overlap to communicate synchronously.

Proposition 9: Time separation reduces the range of coordination mechanisms available to the team – the more time zones are represented in a team the narrower the choice of mechanisms.

Because synchronous communication options are reduced, asynchronous communication and task programming mechanisms become more predominant. Team members still communicate synchronously, but this requires long work hours, widespread schedule shifts¹, and substantial effort planning meetings. The amount of effort required to plan synchronous meetings with team members in multiple

time zones is daunting, therefore tasks are coordinated to a large extent with task programming mechanisms. The most popular task programming mechanism used with time separation, mentioned by the majority of participants is shared documents. The next most often mentioned task programming mechanism is rigorous procedures and routines adhered to with discipline, substantial task organization, well organized workflows, and redundant roles in different sites, coupled with the use of collaboration technologies like Share Point™ and shared drives. Meeting team members face to face before the project starts to get to know each other, get trained on global teamwork, and carefully plan these procedures and routines had a great positive effect on one of the teams operating in multiple time zones. Interestingly, once these task programming coordination mechanisms are in place, they are employed extensively and successfully to coordinate work across all collaboration contexts—i.e., co-located, distant, and time-separated.

Proposition 10: The use of task programming coordination mechanisms increases with time separation – this increase is more substantial when more time zones are represented in a team.

The majority of participants indicated that transactive memory (i.e., knowing who knows what) [37] is most important, when working across time zones. Task awareness (i.e., knowing who has done what and when) was the second most often mentioned cognitive mechanism, followed by presence awareness (i.e., knowing who is around and when, and knowing where and when to find team members). A few members also mentioned that shared knowledge about the task or the team is important with colleagues in other time zones, and a few of these also discussed the importance of shared beliefs (e.g., common vision, shared goals). These cognitive mechanisms were almost as important with members separated by distance only. However, most participants discussed that, while these mechanisms are always important, this knowledge can be more easily acquired when team members are co-located or not widely separated by time.

Proposition 11: The importance of cognitive mechanisms for coordination increases with time separation.

4.6 The Effect of Coordination Processes and Mechanisms on Outcomes (P→O)

Our findings suggest that there is no clear difference in the effectiveness of different coordination mechanisms in helping reduce coordination costs and increase coordination effectiveness. This may sound counterintuitive but it actually suggests something very important and interesting about teamwork and global collaboration across distance and time. It is very telling that the perceived usefulness of

¹ One team shifted hours so much that co-located team members had limited overlapping work time.

coordination mechanisms is so even across the different types of mechanisms (i.e., communication, task programming and cognitive) and within each of these types, with so much variance across participants. Prior research has found that team members have a wide array of mechanisms to coordinate their work, but they will use the mix of mechanisms that fits their preferences and task needs, and that one can substitute the lack of use of one coordination mechanism with the use of another [11]. For example, one team may employ intensive and frequent communication, while another one may employ highly disciplined procedures and division of labor. Yet another team may coordinate implicitly with little communication through cognitive mechanisms. And again, these mechanisms will complement and substitute for each other.

5. Implications and Recommendations to Solution Design(s)

Our findings suggest the following solution design implications and recommendations, to address time separation issues. Note that some of recommendations will need further validation, to be completed in our follow-on research.

5.1 Configurable tools and processes in an integrated team environment

Propositions 8 and 9, and Section 4.6 validate the need for configurable tools and processes to allow global teams to tailor the mechanisms that they use to coordinate their work based on preferences and task needs, while also adjusting dynamically for time overlaps and separation. It may make sense to create initial templates that map to the 5 geographic configurations described in section 4.1 if the patterns of successful coordination differ significantly among the geographic configurations. Our follow-on survey study (see the Concluding Remarks) will help to determine if this is the case. Moreover, an integrated environment is needed that supports multiple modes of collaboration including real time text, voice, and video, asynchronous coordination, persistent group memory and context, and increased awareness of remote team members, to allow the team to dynamically adjust coordination mechanisms throughout the work day.

5.2 Asynchronous tools and explicit work / task coordination mechanisms

Proposition 10 suggests that it may be beneficial to introduce explicit work/task coordination mechanisms and asynchronous tools especially when more time zones are represented in a team. The use of workflow automation to track status and make workflows more explicit may reduce the amount of effort to manually coordinate the work. It

can also reduce the amount of real-time meetings needed to coordinate work and maximize the overlaps in real time because people come to the meeting prepared to discuss key issues and topic areas, rather than using the time to catch up on status. Proposition 3 also affords some interesting implications for solutions design, especially for technical and process dependencies between team members' activities. Explicit representation and tracking of these dependencies may offset the need for extra coordination effort described in Proposition 6.

5.3 Importance of Shared Context and History

Propositions 2, 10 and 11 support both the need for shared context and history about the people on the team, the project goals and objectives, and the actual content that the project team is producing. Solution design implications include a persistent project and team workspace for documenting project assumptions and storing content, but that also includes a visual representation for each member of the team with quick links to their bio, resume, publications, contact information, hobbies, interests, country/culture, and current time of day in their location. This visual representation also provides availability cues and serves as an instant communication launch pad.

5.4 Smart Presence

Propositions 2, 10 and 11 support both the need for shared context and history about the people on the team, the project goals and objectives, and the actual content that the project team is producing. Solution design implications include a persistent project and team workspace for documenting project assumptions and storing content, but that also includes a visual representation for each member of the team with quick links to their bio, resume, publications, contact information, hobbies, interests, country/culture, and current time of day in their location. This visual representation also provides availability cues and serves as an instant communication launch pad.

5.5 Low Hanging Fruit

Until the ultimate collaboration solution becomes available, things that we can do now to improve global team effectiveness include:

- Meet Face to Face before the project starts to help team members get to know each other, establish shared project goals and objectives, assign roles and responsibilities, and agree on team interaction practices that suits team members' needs and preferences (e.g., who can be contacted when and how).
- Agree to and establish coordination procedures and routines up front (at the Face to Face, if possible).

- Establish at least one single point of contact at each site to coordinate with the other sites. This point of contact should adopt a flexible work schedule to increase overlaps with remote team members who are also points of contact for their site.
- Rotate meeting times across geographies so no one geo meets late in their night all of the time. No meetings between midnight and 6 am in any of the participating geo's.
- When the work is ambiguous and/or uncertain, partition and modularize the work so that there are the fewest number of dependencies between remote sites, looking for logical hand-offs whenever possible.

6. Concluding Remarks

In this study we have investigated how various global situational factors affect coordination outcomes in teams, with a focus on the effect of time separation. We have also investigated how time separation influences which mechanisms team members use to coordinate and how the use of these mechanisms affects coordination outcomes. The most salient findings are that: (1) time separation has a strong impact on coordination costs, particularly delay and effort, but this increased effort helps teams achieve coordination success; (2) as more time zones are involved in a project collaboration, the challenges and effort required to achieve success escalate rapidly, thus the importance of understanding how different team configurations across time zones affect team coordination patterns and how teams need to coordinate; (3) coordination mechanisms can be substituted interchangeably, yet there is no clear indication of which coordination mechanism is more effective overall, although the effectiveness of these mechanisms may vary depending on the nature of the task and the task priorities involved; (4) however, co-location, distance and time separation have a substantial impact in the availability of these mechanisms.

These findings underscore the importance of developing a deep understanding of which coordination mechanisms are most effective for the particular tasks that time-separated team members carry out and the mapping of these mechanisms and coordination patterns to different time separated team configurations. One important conclusion of this study is that one of the most important issues affecting coordination in time-separated contexts is the geographical configuration of the team. As more internal boundaries are present in a team (e.g., distance, time zones, culture, language), the availability of coordination mechanisms and processes narrows, and the effort and delay incurred expands. For example, the differences in coordination challenges between a team spanning two time zones and one that spans multiple time zones is substantial.

Consequently, we plan to continue this research with a follow-on survey study (currently underway), which involves: (1) identifying an equal number of teams in each of the 5 configurations discussed in 5.1 above and applying the survey study to as many members as possible in this sample of teams; (2) collecting general background and context data on these team members and their projects; (3) collecting data for each dyad in each team (e.g., time zone separation, communication mechanisms and process used, other coordination mechanisms used, etc.) – this dyadic data will later be aggregated to the team level; (4) collecting team- and project-level data on coordination costs, coordination outcomes and other team and project outcomes of interest. The survey instrument is web-based and dynamic (i.e., scripts pick up data on team membership and ask participants questions about each member in their teams).

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